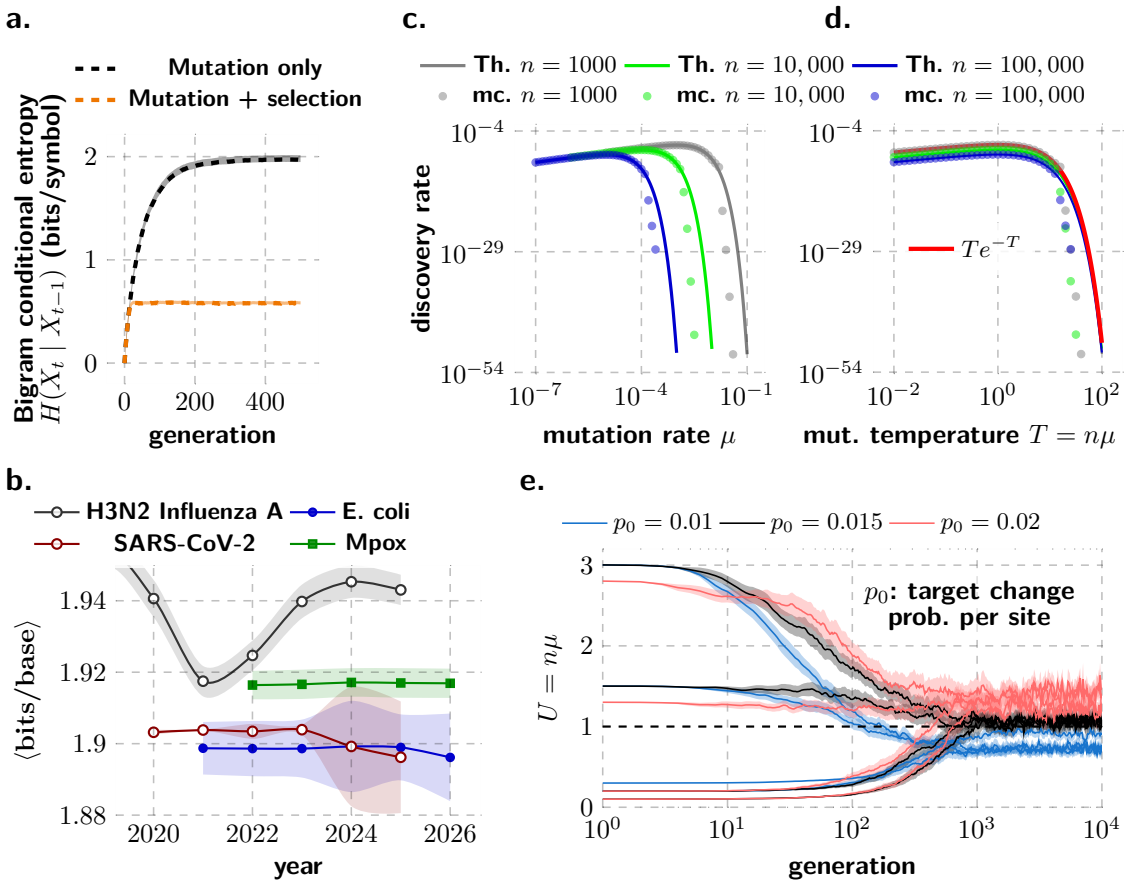


Across organisms and viruses, the per-site mutation rate scales inversely with genome length, yielding an approximately constant number of mutations per genome per generation (Drake's rule). Existing explanations emphasize biochemical fidelity constraints or population-genetic error-threshold arguments that limit mutation from above. Here we provide a complementary, mechanism-independent account based on variation supply. Treating mutation as a blind local perturbation process, we ask which mutation rates maximize the expected rate of producing variants that are atypical relative to the mutation kernel itself, and thus supplied at elevated rate relative to the mutational background, and therefore available for selective amplification. Using counting bounds for the fraction of such mutation-atypical outcomes within Hamming neighborhoods and combining them with a standard local mutation model (e.g., independent per-site substitutions), we obtain an explicit discovery-rate expression that is maximized when the per-genome mutation intensity $T = n\mu$ is $O(1)$, with a peak near $T \approx 1$. Consequently, the optimal per-site rate scales as $\mu^\star = (1 + o(1))/n$. These results derive Drake's rule from a generic supply–destruction tradeoff under local stochastic perturbations, complementing fidelity- and stability-based perspectives.